

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – InterviewAI – Intelligent Interview Trainer Agent

Domain of Project – Education / Career Development

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement - InterviewAI – Intelligent Interview Trainer Agent

The Challenge

Interview preparation is an important part of getting a job, but many students and job seekers struggle with **finding role-specific questions, practicing answers, and receiving proper feedback**. Interview questions are often scattered across different sources, and users do not always know whether their answers are correct or how they can improve.

Without **personalized practice and instant feedback**, candidates may find it difficult to identify their strengths, weaknesses, and areas that require improvement.

The Objective

Build an **Intelligent Interview Trainer Agent** that provides **role-specific interview practice and instant feedback** for students and job seekers. The solution should:

- **Role-Specific Questions** – Generate questions based on selected job roles such as Java Developer, Python Developer, Web Developer, Data Analyst, AI Engineer, and Software Engineer.
- **Experience Level Filtering** – Provide questions suitable for Fresher, Beginner, Intermediate, and Experienced candidates.
- **Interview Categories** – Support Technical, HR, and Behavioral interview questions.
- **Answer Evaluation & Feedback** – Evaluate submitted answers and provide instant scores, strengths, and areas for improvement.
- **Suggested Answers** – Display an ideal answer to help users understand important concepts.
- **Interview Tips** – Provide professional tips to improve interview performance.
- **Progress Dashboard** – Track questions attempted, average score, job role, experience level, and question history.

Proposed Solution - InterviewAI – Intelligent Interview Trainer Agent

The proposed solution is an **Intelligent Interview Trainer Agent** designed to help students and job seekers prepare for interviews through role-specific questions, answer evaluation, and instant feedback.

The system provides a structured interview practice experience based on the user's selected job role, experience level, and interview category.

The solution includes:

- **Interview Setup Module** – Collects user details such as name, job role, experience level, and interview category.
- **Question Generation Module** – Selects and displays relevant questions based on the user's chosen profile.
- **Role-Specific Question Database** – Contains questions for Java Developer, Python Developer, Web Developer, Data Analyst, AI Engineer, and Software Engineer.
- **Answer Evaluation Module** – Evaluates answers using keyword matching, answer length, and STAR-method vocabulary.
- **Feedback Module** – Provides instant scores, strengths, areas for improvement, suggested answers, and interview tips.
- **Question Navigation** – Allows users to submit answers, skip questions, and continue to the next question.
- **Progress Dashboard** – Displays questions attempted, average score, job role, experience level, and question history.

Overall, **InterviewAI provides an interactive and structured platform for interview preparation**, helping users practice regularly, evaluate their answers, identify improvement areas, and improve their interview performance.

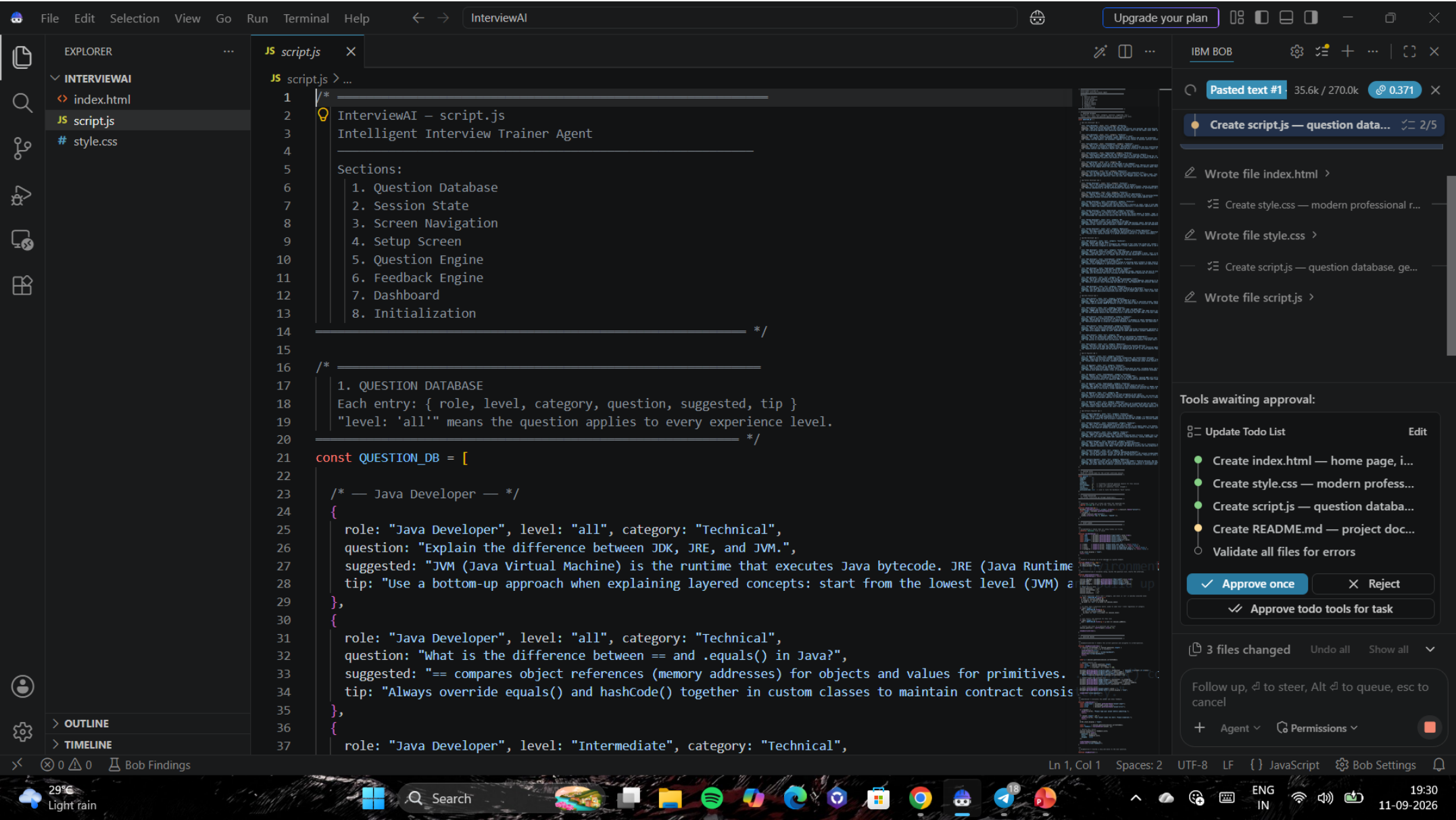
Technology Used

- **IBM Bob Platform** – Used for AI-assisted project development, code generation, file creation, and implementation workflow.
- **HTML5** – Used to create the structure of the InterviewAI web application, including Home, Setup, Question, Feedback, and Dashboard screens.
- **CSS3** – Used for designing the user interface, responsive layout, cards, forms, buttons, progress bars, and overall application styling.
- **JavaScript (ES6+)** – Used for implementing the question database, question filtering, scoring engine, feedback system, navigation, and dashboard functionality.
- **GitHub** – Used for version control and hosting the public project repository.
- **Web Browser / Live Server** – Used for running and testing the InterviewAI application locally.

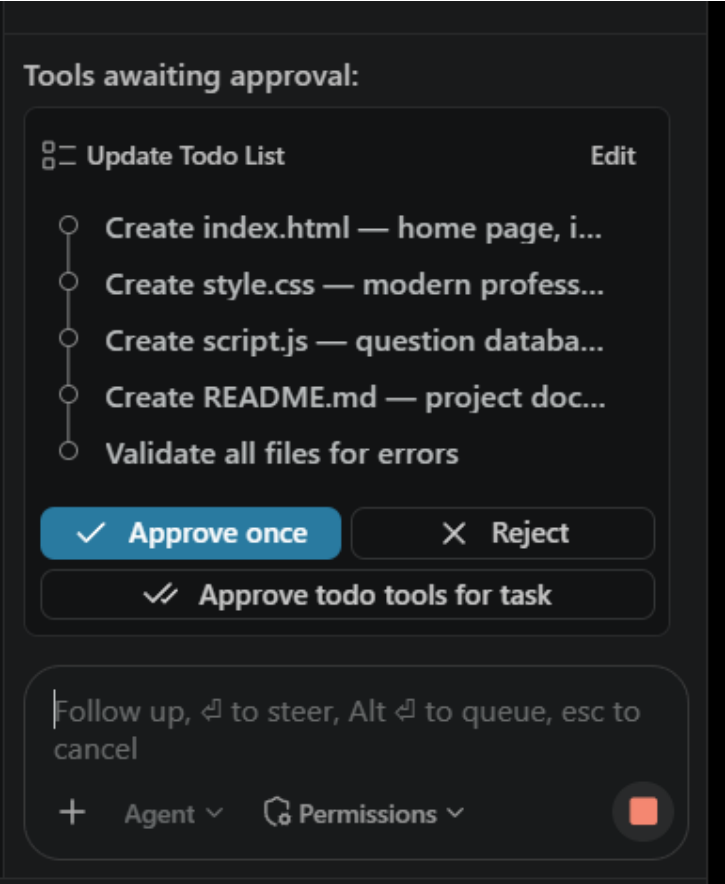
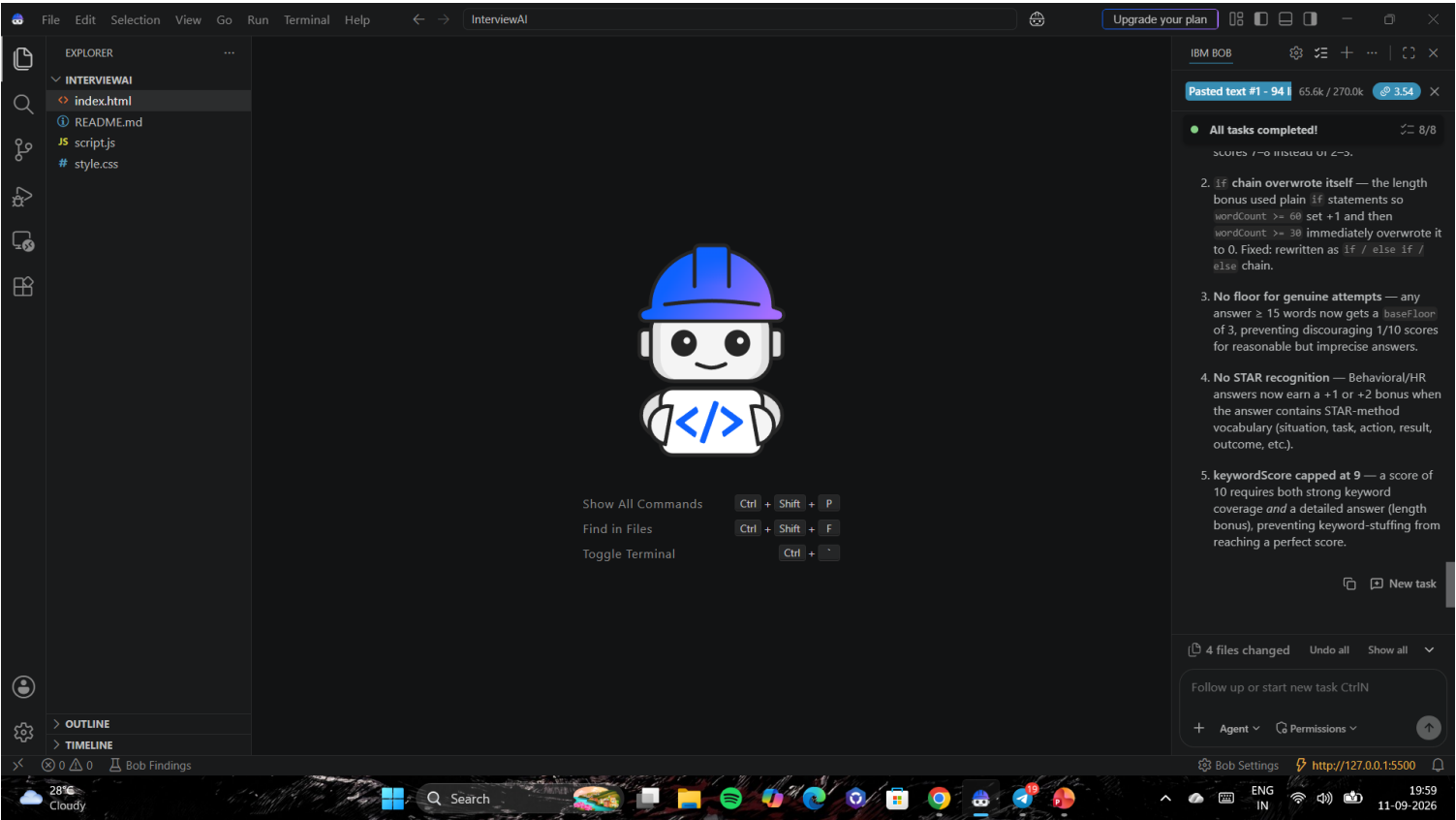
Development Approach

The project is a **complete client-side web application** with no external framework or backend dependency. All interview questions, scoring logic, feedback generation, and session tracking are handled using **Vanilla JavaScript**.

Required Files created using IBM bob -



IBM Bob Screenshots:



FileEditSelectionViewGoRunTerminalHelp

InterviewAI

Upgrade your plan

EXPLORER

INTERVIEWAI

New folder

index.html

README.md

script.js

style.css

OUTLINE

TIMELINE

index.html

index.html > html > body

```
2 <html lang="en">
9 <body>
14 <div id="screen-home" class="screen active">
29 <section class="features">
30 <div class="container">
41 <p>Receive a score out of 10, identified strengths, areas for improvement, and
42 </div>
43 <div class="feature-card">
44 <div class="feature-icon">&#128203;</div>
45 <h3>Progress Dashboard</h3>
46 <p>Track total questions attempted, your average score, and session details at a glance.
47 </div>
48 <div class="feature-card">
49 <div class="feature-icon">&#128161;</div>
50 <h3>Pro Interview Tips</h3>
51 <p>Each feedback response includes a professional tip to sharpen your interview skills.
52 </div>
53 </div>
54 </div>
55 </section>
56
57 <!-- How it works -->
58 <section class="how-it-works">
59 <div class="container">
60 <h2 class="section-title">How It Works</h2>
61 <div class="steps">
62 <div class="step">
63 <div class="step-number">1</div>
64 <div class="step-text">
65 <h4>Set Up Your Profile</h4>
66 <p>Enter your name, target job role, experience level, and interview category.</p>
67 </div>
68 </div>
69 <div class="step">
70 <div class="step-number">2</div>
71 <div class="step-text">
72 <h4>Answer Questions</h4>
```

IBM BOB

86%

Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

Users should independently verify accuracy of AI-generated content.

Recent Tasks

Create a complete web application nam...

1 hr ago 6.44

Build Anything, @ for context, / for commands, ⬆ drag to attach

+ Agent Permissions

main

0 0

Bob Findings

Varun Kushwah (1 hour ago) Ln 11, Col 55 Spaces: 2 UTF-8 LF HTML Bob Settings Go Live

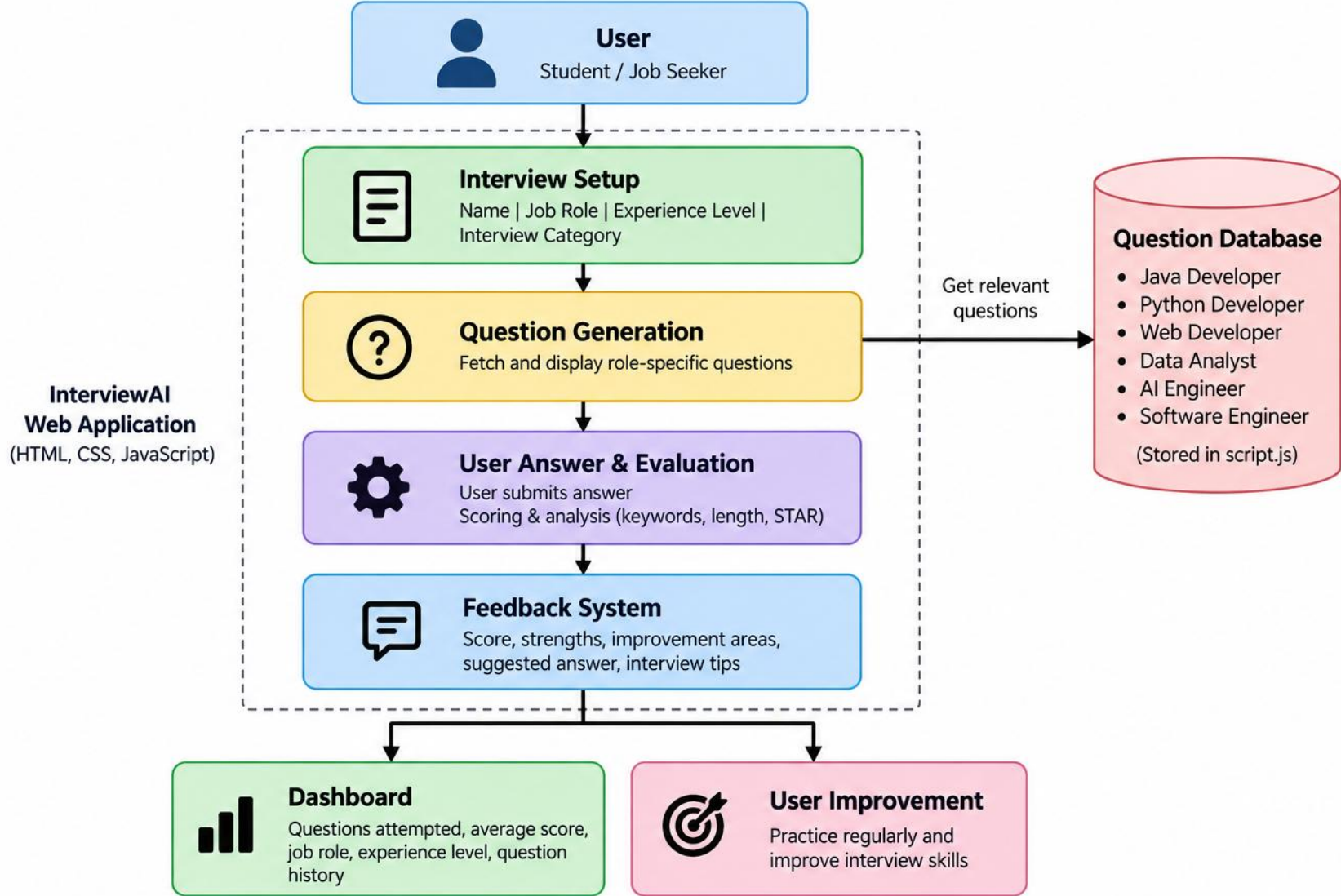
27°C Mostly cloudy

Search

21

21:57 11-09-2026

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Role of AI in InterviewAI

InterviewAI uses an **AI-style intelligent evaluation system** to provide personalized interview practice and instant feedback.

The system analyzes the user's answer based on:

Keyword Matching – Compares the answer with important concepts from the ideal answer.

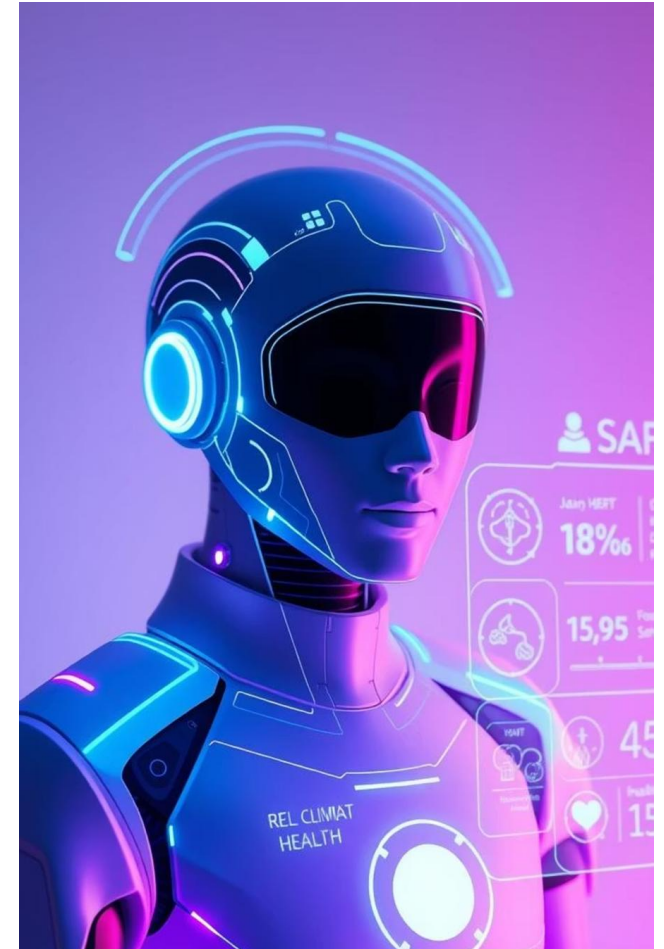
Answer Length Analysis – Evaluates whether the response contains sufficient detail.

STAR Method Recognition – Identifies Situation, Task, Action, and Result concepts for HR and Behavioral questions.

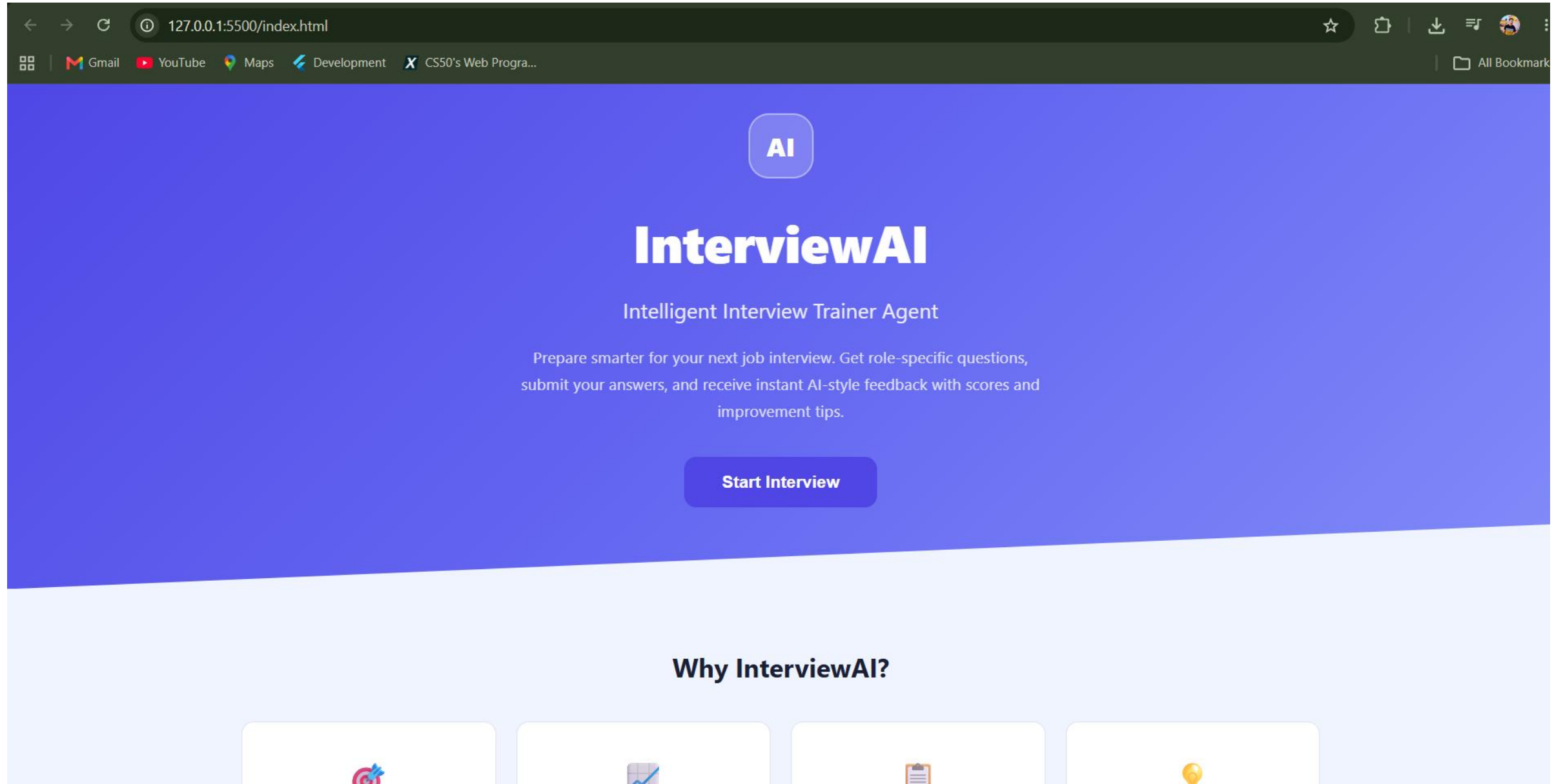
Instant Feedback – Provides a score, strengths, areas for improvement, suggested answer, and interview tips.

Personalized Practice – Generates questions based on selected job role, experience level, and interview category.

Overall, the intelligent evaluation system makes InterviewAI a **personalized interview training platform** that helps users identify weaknesses and improve their interview performance.



Project Output Screenshot



Project Output Screenshot



[← Home](#)

Interview Setup

Tell Us About Yourself

We'll generate questions perfectly matched to your profile.

Your Name

Varun Kushwah

Job Role

Java Developer

Experience Level

Fresher

Interview Category

Technical

Generate Questions

Project Output Screenshot

← → ↻ ⓘ 127.0.0.1:5500/index.html 🔍 ☆ 📌 ⬇️ 📑 👤 ⋮

🗖️ | 📧 Gmail 📺 YouTube 📍 Maps ⚡ Development 📄 CS50's Web Progra...

📁 All Bookmarks

← Setup Java Developer Dashboard

Question 1 of 4 Score: 0/0

TECHNICAL FRESHER

Explain the difference between JDK, JRE, and JVM.

Your Answer

development tools such as the compiler.

JRE (Java Runtime Environment) provides the libraries and environment required to run Java applications.

JVM (Java Virtual Machine) executes Java bytecode and allows Java programs to run on different operating systems, providing platform independence.

Submit Answer Skip Question

Project Output Screenshot

Hello, Varun Kushwah!

You've completed 4 questions so far.

4

QUESTIONS ATTEMPTED

8.5

AVERAGE SCORE

Java

Developer

JOB ROLE

Fresher

EXPERIENCE LEVEL

Question History

1. What is the difference between ArrayList and LinkedList in Java?

4/10

2. What is the difference between == and .equals() in Java?

10/10

3. Explain the difference between JDK, JRE, and JVM.

10/10

4. Explain Java's memory model and the significance of the 'volatile' keyword.

10/10

Continue Interview

New Interview



Excellent Answer!

✓ Strengths

Excellent answer! You demonstrated strong coverage of the topic (15/11 target concepts matched) and gave a very detailed response (52 words). You touched on 15 key concepts.

▲ Areas for Improvement

Great job! For perfection, consider adding a specific example or analogy to make your answer more memorable for the interviewer.

 Suggested Answer

JVM (Java Virtual Machine) is the runtime that executes Java bytecode. JRE (Java Runtime Environment) includes JVM plus standard libraries needed to run Java programs. JDK (Java Development Kit) includes JRE plus development tools like the compiler (javac). Developers need the JDK to write and compile code, whereas end-users only need the JRE.

 Interview Tip

Use a bottom-up approach when explaining layered concepts: start from the lowest level (JVM) and build up to the highest (JDK).

Next Question

End Interview

Novelty and Uniqueness

InterviewAI stands out by combining **role-specific interview practice, intelligent answer evaluation, and instant personalized feedback** in a single web application.

Role-Based Interview Preparation – Generates questions tailored to different job roles such as Java Developer, Python Developer, Web Developer, Data Analyst, AI Engineer, and Software Engineer.

Experience-Level Filtering – Adapts questions based on Fresher, Beginner, Intermediate, and Experienced levels.

Intelligent Answer Evaluation – Evaluates answers using keyword matching, answer length analysis, and relevant concept coverage.

STAR Method Recognition – Encourages structured responses for HR and Behavioral interview questions.

Instant Personalized Feedback – Provides a score, strengths, areas for improvement, suggested answer, and professional interview tips.

Progress Tracking – Maintains question history, attempted questions, and average scores through a dashboard.

This makes InterviewAI a **simple, personalized, and interactive interview preparation platform**, rather than just a static collection of interview questions.

Git Hub Link

GitHub Repository Link - <https://github.com/royalvarun15/InterviewAI-AICTE-2026.git>

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Future Scope

1. Integration with Advanced AI Models –

The system can be integrated with advanced AI models to provide more accurate answer evaluation, personalized feedback, and natural interview conversations.

2. Voice-Based Mock Interviews –

A voice interaction feature can allow users to answer interview questions verbally, creating a more realistic interview experience.

3. More Job Roles and Question Categories –

The application can be expanded to support additional job roles, industries, programming languages, and interview categories.

4. Performance Analytics –

Future versions can provide detailed analytics to identify weak areas, track improvement, and recommend personalized practice sessions.

5. User Accounts and Cloud Storage –

User accounts can be added to save interview history, scores, progress, and performance data across multiple sessions.

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Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Varun Kushwah

for the completion of

Getting Started with Artificial Intelligence

(PLAN-E624C2604060)

According to the Your Learning Builder - Plans system of record

Completion date: 11 Sep 2026 (GMT)

Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Varun Kushwah

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 11 Sep 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.

Student Name: Varun Kushwah

Problem Statement No.: 22

Problem Statement: Interview Trainer Agent

Project Name: InterviewAI – Intelligent Interview Trainer Agent

GitHub Repository: <https://github.com/royalvarun15/InterviewAI-AICTE-2026.git>

AICTE Student ID: STU6a64e3ed3219b1784996845